

JEE MAIN REVISION MASTER: BINOMIAL THEOREM

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	General Term, Middle Term & Greatest Term	9 PYQs	Expansion Terms
Standard	Properties & Series Sum of Binomial Coefficients	0 PYQs	Coefficient Identities

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
General Term, Middle Term & Greatest Term	$(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$, $T_{r+1} = {}^nC_r a^{n-r} b^r$, $T_{\frac{n}{2} + 1}$ (if n is even)
Properties & Series Sum of Binomial Coefficients	$\sum_{r=0}^n {}^nC_r = 2^n$, $C_0 - C_1 + C_2 - \dots = 0$, $\sum_{r=1}^n r {}^nC_r = n 2^{n-1}$, $\sum_{r=0}^n ({}^nC_r)^2 = {}^{2n}C_n$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 General Term, Middle Term & Greatest Term (High Yield — 9 PYQs)

Core Principle: Expansion of $(x + a)^n$, formula for general $(r+1)^{\text{th}}$ term, finding term independent of x , and middle term determination.

■ **Caution / JEE Pitfall:** In $(a + b)^n$, total number of terms is $n + 1$. If n is odd, there are TWO middle terms: $(\frac{n+1}{2})^{\text{th}}$ and $(\frac{n+3}{2})^{\text{th}}$.

■ **Speed Trick:** In expansion of $(ax^p + bx^q)^n$, the term independent of x occurs at index $r = (n p / p + q)$ (provided r is an integer).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value at $x=2$ for $\frac{1}{x^3} + 3x^2 + 3x + 1x^4 + 4x^3 + 6x^2 + 4x + 1 \dots$ " [Ans: C]
- [2025 Shift] Q#2 (MCQ): "If the number of terms in the binomial expansion of $(2x+3)^{3n}$ is 22, then the value of n is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The value of ${}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3 + {}^{46}C_3 + {}^{45}C_3 + {}^{45}C_4$ is..." [Ans: N/A]

3.2 Properties & Series Sum of Binomial Coefficients (Standard Yield — 0 PYQs)

Core Principle: Combinatorial identities, summing coefficient series via differentiation and integration of $(1+x)^n$.

■ **Caution / JEE Pitfall:** To find the sum of all coefficients of a polynomial expansion $P(x)$, simply substitute $x = 1$.

■ **Speed Trick:** Useful ratio identity: $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The value at $x=2$ for $\frac{x^3+3x^2+3x+1}{x^4+4x^3+6x^2+4x+1}$..." **[Ans: C]**
- **[2025 Shift] Q#2 (MCQ):** "If the number of terms in the binomial expansion of $(2x+3)^{3n}$ is 22, then the value of n is..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The value at $x=2$ for $\frac{x^3+3x^2+3x+1}{x^4+4x^3+6x^2+4x+1}$

- (A) 3
- (B) $(25 / 61)$
- (C) $(1 / 3)$ **[Correct]**
- (D) $(19 / 73)$

Correct Answer: C

Solution Analysis: The given expression is $\frac{x^3+3x^2+3x+1}{x^4+4x^3+6x^2+4x+1}$. We can recognize both numerator and denominator using identities. $x^3+3x^2+3x+1=(x+1)^3$ and $x^4+4x^3+6x^2+4x+1=(x+1)^4$. So the expression becomes $\frac{(x+1)^3}{(x+1)^4}$. Cancel the common factor $(x+1)^3$: $\frac{(x+1)^3}{(x+1)^4} = \frac{1}{x+1}$. Now put $x=2$: $\frac{1}{2+1} = \frac{1}{3}$. So the correct option is $(1 / 3)$ Option C.